

Spatial Analysis and Topological Relationships in Database

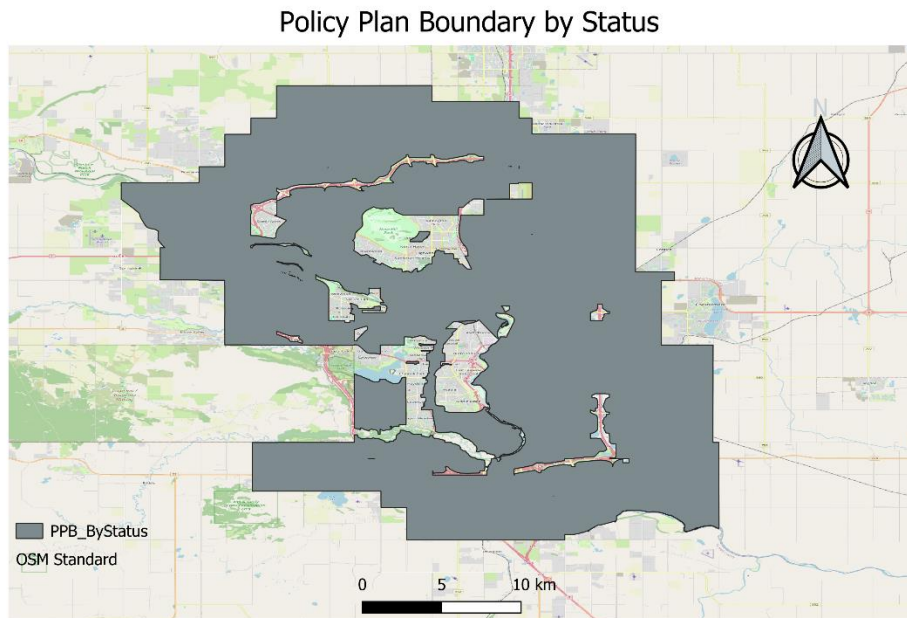
Lab Teaching Assistants:
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Chalotra**

Assignment date:
15/10/2025

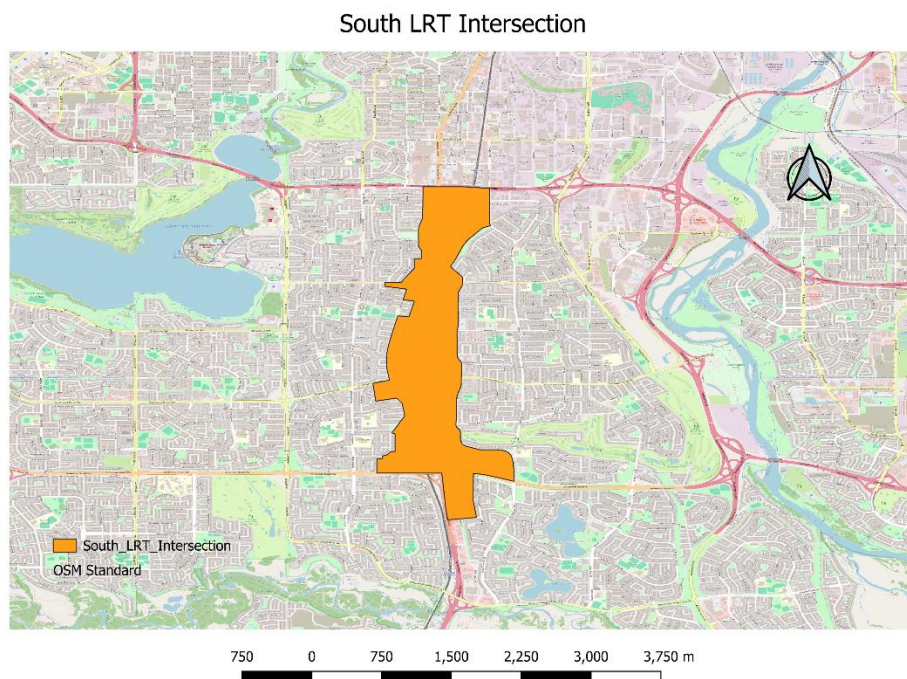
Submission date:
29/10/2025

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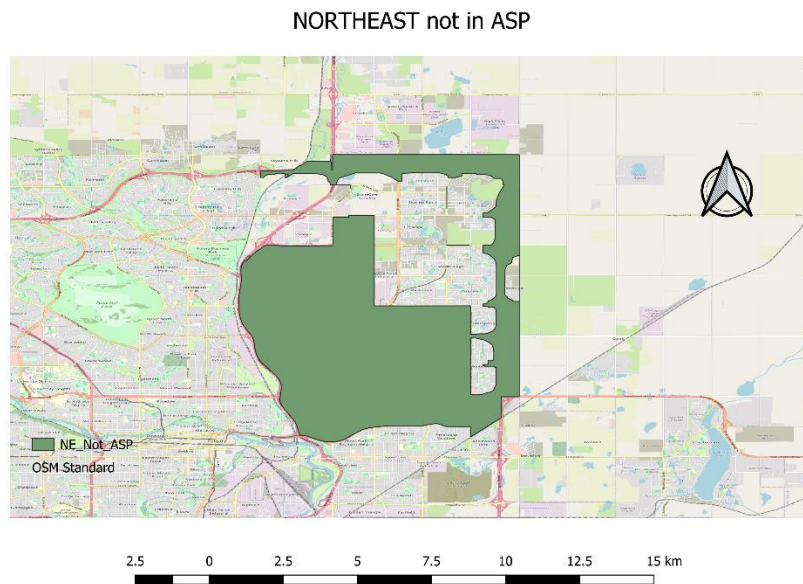
Q1. Join features in “Policy Plan Boundaries” based on their “STATUS”.



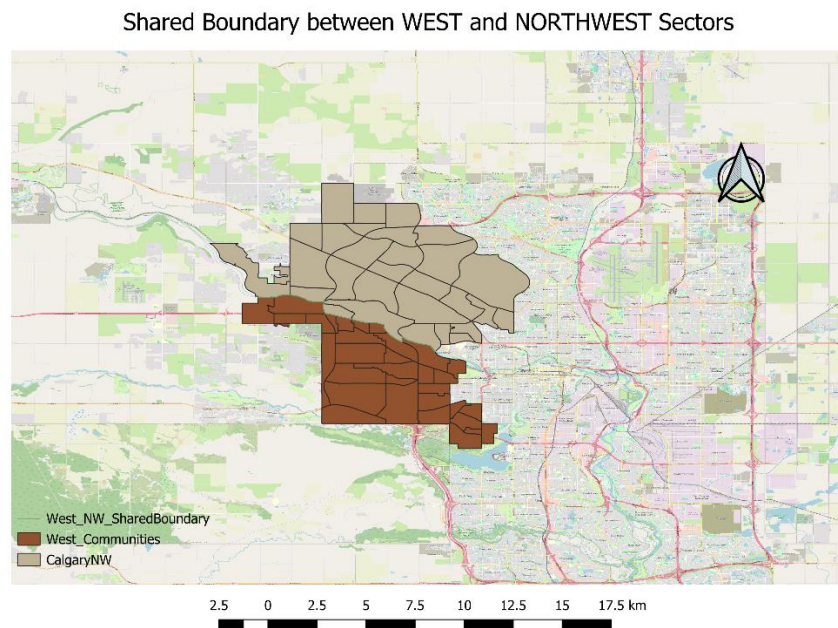
Q2. Find the intersection of “SOUTH” sector with “L.R.T. Policy” plan type of “Policy Plan Boundaries”.



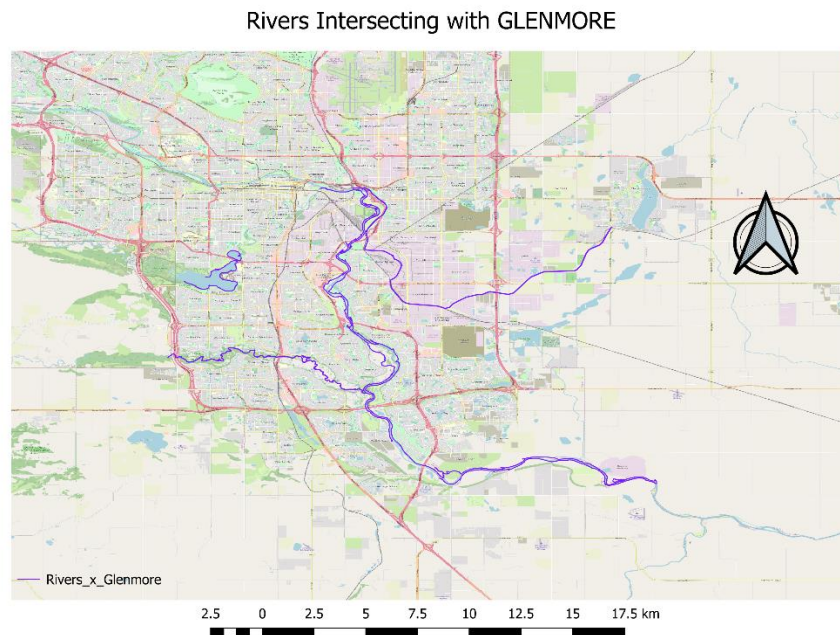
Q3. Find areas in “NORTHEAST” that are not in “Area Structure Plan” type.



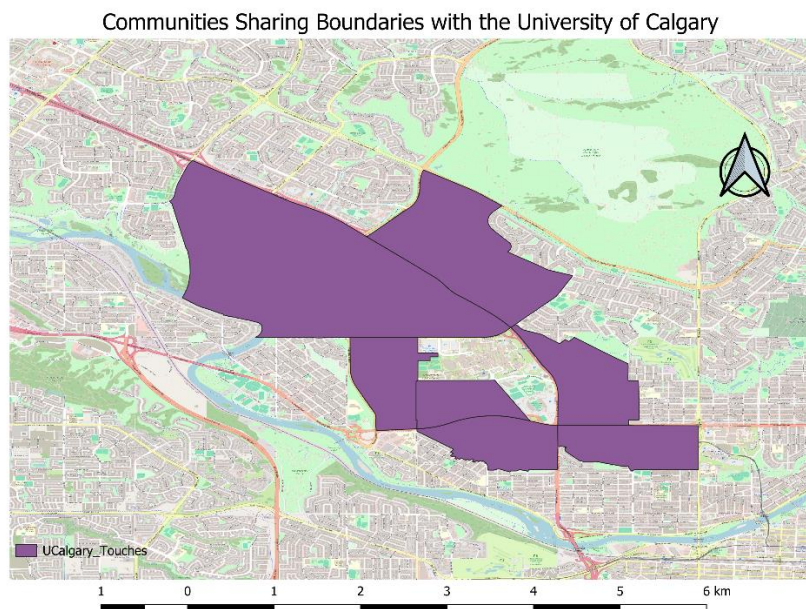
Q4. Find the common boundary between “WEST” and “NORTHWEST” sectors.



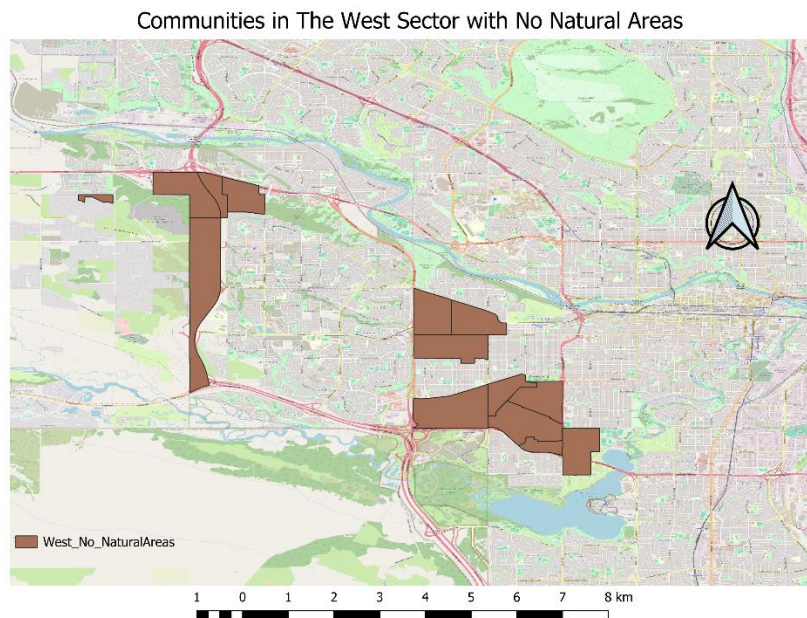
Q5. Find rivers that intersect with “GLENMORE” road. (Do NOT use original Major Road Network. Union features in “Major Road Network” based on their “Name”).



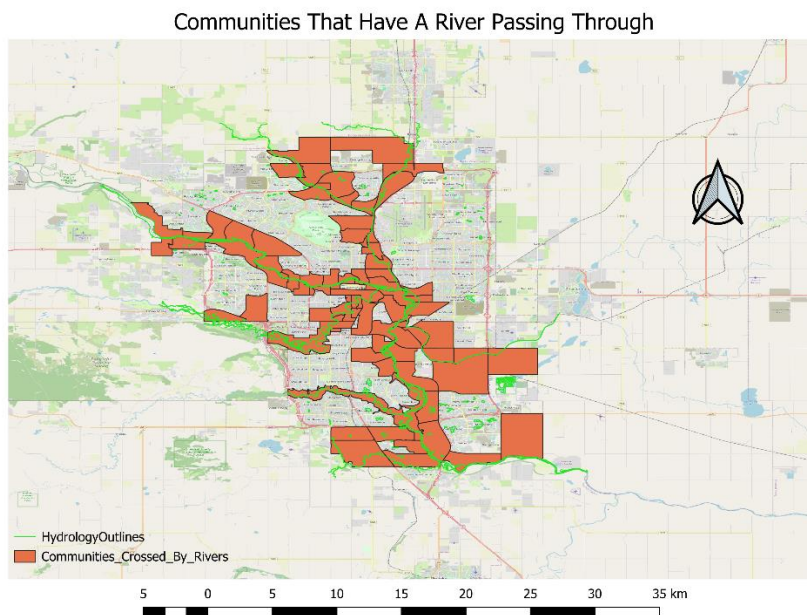
Q6. Find all communities that share their boundaries with “UNIVERSITY OF CALGARY” community. (Use Touch)



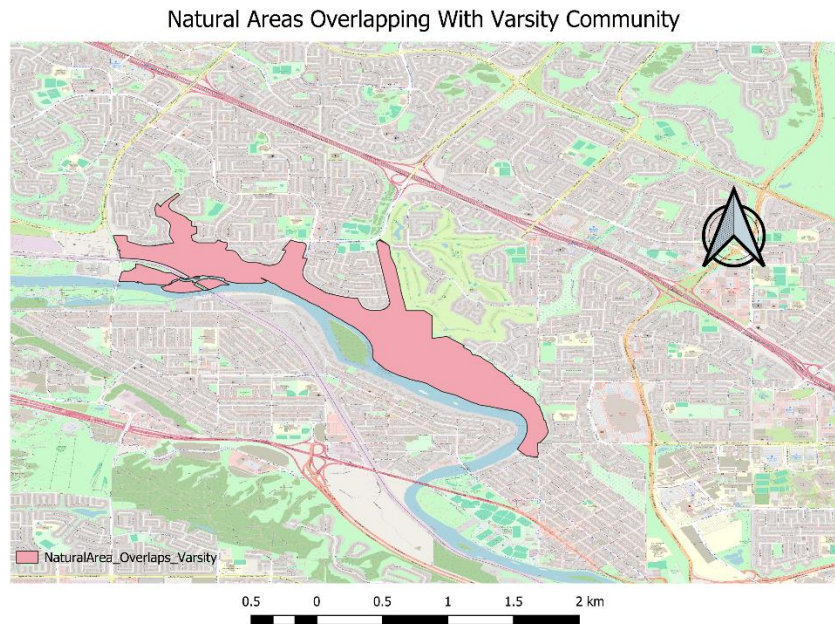
Q7. Find all communities in the “WEST” sector that do not contain any natural area. (Use Disjoint).



Q8. Find all communities that a river passes through. (Use Cross)



Q9. Find the natural area that overlaps with “VARSITY” community.



Q10. Find the topological relationship between natural area with Asset_CD = “SIL245” and “Silver Springs” community. (DE-9IM matrix and its interpretation in topological terms are required).

Showing rows: 1 to 1 Page No: 1 of 1								
	de9im_matrix text	intersects boolean	touches boolean	overlaps boolean	na_within_ss boolean	ss_contains_na boolean	crosses boolean	disjoint boolean
1	212101212	true	false	true	false	false	false	false

Topological Relationship between SIL245 and Silver Springs

Predicate / Function	Meaning	Result
ST_Relate	DE-9IM matrix pattern describing spatial relationship	212101212
ST_Intersects	<i>TRUE</i> if geometries share any portion of space	TRUE
ST_Touches	<i>TRUE</i> if boundaries meet but interiors do not intersect	FALSE
ST_Overlaps	<i>TRUE</i> if interiors partially overlap but neither completely contains the other	TRUE
ST_Within	<i>TRUE</i> if SIL245 is completely inside Silver Springs	FALSE
ST_Contains	<i>TRUE</i> if Silver Springs completely contains SIL245	FALSE
ST_Crosses	<i>TRUE</i> if geometries intersect but have different dimensional overlap	FALSE
ST_Disjoint	<i>TRUE</i> if geometries share no common space	FALSE

Interpretation

The DE-9IM matrix (212101212) and associated Boolean results indicate that the **SIL245** natural area and the **Silver Springs** community share a portion of overlapping space without either polygon completely containing the other. The positive results for ST_Intersects and ST_Overlaps, combined with negative results for ST_Within and ST_Contains, confirm that this is a **partial overlap relationship**.

In topological terms, the two polygons' interiors intersect, and their boundaries overlap in some areas, but parts of each geometry also extend beyond the other. This means **SIL245 and Silver Springs overlap spatially**, sharing common interior regions but remaining distinct geographic entities.